

Zachary Nason

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EDUCATION

B.S., Mathematics 2020
University of Kansas, Lawrence, KS

M.S., Mathematics 2023
University of Nebraska-Lincoln, Lincoln, NE

Ph.D., Mathematics Expected May 2027
University of Nebraska-Lincoln, Lincoln, NE
Advisor: Thomas Marley

POSITIONS HELD

Graduate Teaching Assistant August 2021 - Present
Mathematics Department, University of Nebraska-Lincoln

Research Training Group in the Mathematical Sciences (RTG) Graduate Trainee Fall 2024
Mathematics Department, University of Nebraska-Lincoln

AWARDS

Emeritus Faculty Fellowship - Lloyd Jackson Award 2025
This award is given by the UNL Mathematics Department to graduate students to recognize notable scholarship and to promote future mathematical research. This award is named after UNL emeritus professor Lloyd Jackson.

Outstanding Qualifying Exam 2022
This award is given by the UNL Mathematics Department to the graduate student who has achieved the highest scores on the Ph.D. qualifying exams for that year.

PUBLICATIONS/PREPRINTS

- [1] Luigi Ferraro and Zachary Nason. Width over DG-rings. In preparation.
- [2] Luigi Ferraro and Zachary Nason. Intersection theorems over DG-rings revisited. Accessible at <https://arxiv.org/abs/2606.32031> (submitted), June 2026.
- [3] Zachary Nason, Andrew J. Soto Levins, and Ryan Watson. Quasi-Gorenstein morphisms of commutative local dg-algebras. Accessible at <https://arxiv.org/abs/2605.02029> (submitted), May 2026.
- [4] Lars Winther Christensen, Antonia Kekkou, Justin Lyle, Zachary Nason, and Andrew J. Soto Levins. G-levels of perfect complexes and a Bass formula for levels. Accessible at <https://arxiv.org/abs/2507.10486> (submitted), April 2026.
- [5] Paulo Martins, Victor D. Mendoza Rubio, and Zachary Nason. Finiteness of complete intersection dimensions of RHom complexes. Accessible at <https://arxiv.org/abs/2601.07811> (submitted), January 2026.
- [6] Zachary Nason. Maximal Cohen-Macaulay DG-complexes. *Journal of Pure and Applied Algebra*, 230(2):108193, 2026.

TALKS AND PRESENTATIONS

EXTERNAL PRESENTATIONS

7. **Quasi-Gorenstein morphisms of DG-rings and exact sequences** (20 min) January 12, 2027
Joint Mathematics Meetings 2027, Chicago
Special Session on Homological Methods, Syzygies, and Resolutions in Commutative Algebra
6. **Quasi-Gorenstein morphisms of DG-rings and exact sequences** (20 min) October 10, 2026
KUMUNU, University of Kansas
5. **Maximal Cohen-Macaulay DG-complexes** (20 min) August 12, 2026
Pontifical Catholic University of Peru, XXV Latin American Colloquium on Algebra
4. **Maximal Cohen-Macaulay DG-complexes** (50 min) April 13, 2026
University of Texas-Rio Grande Valley, Algebra Seminar
3. **Level inequalities for complexes** (50 min) March 4, 2026
Texas Tech University, Algebra and Number Theory Seminar
2. **Level inequalities for complexes** (50 min) January 30, 2026
Syracuse University, Algebra Seminar
1. **Maximal Cohen-Macaulay DG-complexes and a derived Improved New Intersection Theorem** (20 min) March 29, 2025
AMS Spring Central Sectional 2025, University of Kansas
Special Session on Topics in Commutative Algebra

INTERNAL PRESENTATIONS

13. **Quasi-Gorenstein morphisms of DG-rings and exact sequences** (50 min) September 2, 2026
University of Nebraska-Lincoln, Commutative Algebra Reading Seminar
12. **Quasi-Gorenstein morphisms of DG-rings and the virtually G-small property** (50 min) August 30, 2026
University of Nebraska-Lincoln, URiCA 2026
11. **Koszul property for points in projective space** (50 min) April 6, 2026
University of Nebraska-Lincoln, RTG Learning Seminar: Koszul Algebras
10. **Level inequalities for complexes** (50 min) December 3, 2025
University of Nebraska-Lincoln, Commutative Algebra Seminar
9. **Journeying towards Jensen's theorem** (3 talks, 50 min each) July 21, 23, and 25, 2025
University of Nebraska-Lincoln, Summer Commutative Algebra Reading Seminar
8. **Maximal Cohen-Macaulay DG-modules** (50 min) November 20, 2024
University of Nebraska-Lincoln, Commutative Algebra Seminar
7. **Maximal Cohen-Macaulay complexes and an extension to differential graded rings** (50 min) September 25, 2024
University of Nebraska-Lincoln, Commutative Algebra Reading Seminar
6. **Constructing the derived category** (3 talks, 50 min each) July 29, 31, and August 2, 2024
University of Nebraska-Lincoln, Summer Commutative Algebra Reading Seminar
5. **Cohen-Macaulay differential graded algebras** (50 min) May 1, 2024
University of Nebraska-Lincoln, Commutative Algebra Reading Seminar
4. **Gorenstein differential graded algebras** (50 min) October 19, 2023
University of Nebraska-Lincoln, Commutative Algebra Reading Seminar

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| 3. Characterizing Gorenstein local rings using the Frobenius map (50 min)
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | April 19, 2023 |
| 2. Symbolic powers and monomial ideals (50 min)
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | November 16, 2022 |
| 1. A counterexample concerning the existence of height one prime ideals in the intersection of height two prime ideals (50 min)
<i>University of Nebraska-Lincoln, Commutative Algebra Reading Seminar</i> | February 9, 2022 |

CONFERENCES ATTENDED

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| • Arbeitsgemeinschaft on New Homological Methods in Toric Geometry
<i>Mathematisches Forschungsinstitut Oberwolfach, Germany</i> | October 2026 |
| • KUMUNU
<i>University of Kansas</i> | October 2026 |
| • URiCA
<i>University of Nebraska-Lincoln</i> | August 2026 |
| • XXV Latin American Colloquium on Algebra
<i>Pontifical Catholic University of Peru, Peru</i> | August 2026 |
| • A Celebration of Golod's Mathematical Legacy
<i>University of Warwick, UK</i> | April 2026 |
| • CA+ 2026
<i>Iowa State University</i> | March 2026 |
| • SLAM 2026
<i>The University of Texas at Arlington</i> | March 2026 |
| • KUMUNU
<i>University of Missouri</i> | October 2025 |
| • AMS Fall 2025 Southeastern Sectional Meeting
<i>Tulane University</i> | October 2025 |
| • Summer CAMP
<i>University of Nebraska-Lincoln</i> | August 2025 |
| • Modules and Rings: Recent Developments in Commutative Algebra
<i>University of Genoa, Italy</i> | June 2025 |
| • URiCA
<i>University of Nebraska-Lincoln</i> | May 2025 |
| • RayFest
<i>University of Nebraska-Lincoln</i> | April 2025 |
| • AMS Spring 2025 Central Sectional Meeting
<i>University of Kansas</i> | March 2025 |
| • Commutative Algebra in Mixed Characteristic
<i>University of Nebraska-Lincoln</i> | March 2025 |
| • CA+ 2025
<i>Iowa State University</i> | March 2025 |

- KUMUNU
University of Missouri September 2024
- URiCA
University of Nebraska-Lincoln May 2024
- KUMUNU
University of Missouri October 2023
- AMS Fall 2023 Central Sectional Meeting
Creighton University September 2023
- BrianFest
University of Nebraska-Lincoln August 2023
- URiCA
University of Nebraska-Lincoln May 2023
- Morgantown Algebra Days
West Virginia University April 2023
- Math 125
University of Nebraska-Lincoln April 2023
- KUMUNU
University of Nebraska-Lincoln October 2022

TEACHING

Instructor of Record MATH 315: <i>Linear Algebra for Data Science</i>	Fall 2026
Recitation Instructor MATH 221: <i>Differential Equations</i>	Fall 2026 (2 sections)
Recitation Instructor MATH 221: <i>Differential Equations</i>	Spring 2026 (2 sections)
Grading Assignment MATH 918: <i>Topics in Algebra - Positive Characteristic Methods in Commutative Algebra</i>	Spring 2026
Instructor of Record MATH 103: <i>College Algebra and Trigonometry</i>	Fall 2025
Grading Assignment MATH 915: <i>Homological Algebra</i>	Fall 2025
Instructor of Record MATH 107WHT: <i>Calculus II - William H. Thompson Scholars Section</i>	Spring 2025
Instructor of Record MATH 208: <i>Calculus III</i>	Summer 2024
Instructor of Record MATH 104: <i>Applied Calculus</i>	Spring 2024
Instructor of Record MATH 103: <i>College Algebra and Trigonometry</i>	Fall 2023
Grading Assignment MATH 825: <i>Mathematical Analysis I</i>	Fall 2023

Instructor of Record MATH 107: <i>Calculus II</i>	Summer 2023
Algebra Qualifying Exam Review Session Organizer Organized and led the review session for UNL's algebra qualifying exam.	
Recitation Instructor MATH 208: <i>Calculus III</i>	Spring 2023 (3 sections)
Instructor of Record MATH 101: <i>College Algebra</i>	Fall 2022
Recitation Instructor MATH 107: <i>Calculus II</i>	Spring 2022 (2 sections), Summer 2022 (1 section)
Recitation Instructor MATH 106: <i>Calculus I</i>	Fall 2021
Tutoring Drop-in math tutoring center for UNL first-year undergraduates. Tutored for two hours per week, one semester per year.	

SERVICE

Commutative Algebra Reading Seminar Co-Organizer <i>University of Nebraska-Lincoln</i> Organized and led a weekly graduate student seminar in commutative algebra along with two other graduate students.	2025-Present
UNL DRP Mentor <i>University of Nebraska-Lincoln</i> Mentored an advanced undergraduate student in basic commutative algebra and planar algebraic geometry, with the eventual goal of proving Bezout's Theorem for projective curves in projective 2-space	2024-2025
UNL Graduate Mentor <i>University of Nebraska-Lincoln</i> Mentored a beginning mathematics graduate student during his first year of graduate school. Advised and helped the first-year graduate students with navigating academic life and with adjusting to Lincoln.	2023-2024
UNL Math Day Volunteer <i>University of Nebraska-Lincoln</i> The mathematics department at the University of Nebraska-Lincoln holds an annual math day for high school students in Nebraska. Students compete in a math bowl and participate in mathematical activities and puzzle-solving.	2021-Present

COMPUTER SKILLS

Languages: Java, Python, C++

Software: $\text{\LaTeX}$, Macaulay2, MATLAB, Microsoft Office, Google Workspace